RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:19 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:20 AM
Time taken	44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs(1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

For example:

Input	Result
3 1 2 3 4 5 6	28

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Comparison function for ascending order
int compare_asc(const void *a, const void *b) {
    int val1 = *(const int *)a;
    int val2 = *(const int *)b;
    if (val1 < val2) return -1;
    if (val1 > val2) return 1;
    return 0;
}

// Comparison function for descending order
int compare_desc(const void *a, const void *b) {
    int val1 = *(const int *)a;
    int val2 = *(const int *)b;
    if (val1 > val2) return -1;
    if (val1 < val2) return 1;
    return 0;
}

int main() {
    int n;

    // Read the size of the arrays
    if (scanf("%d", &n) != 1) {
        return 0;
    }

    // Dynamically allocate memory for both arrays
    int *array_One = (int *)malloc(n * sizeof(int));
    int *array_Two = (int *)malloc(n * sizeof(int));

    if (array_One == NULL || array_Two == NULL) {
        return 1; // Memory allocation failed
    }

    // Read elements for array_One
    for (int i = 0; i < n; i++) {
        scanf("%d", &array_One[i]);
    }

    // Read elements for array_Two
    for (int i = 0; i < n; i++) {
        scanf("%d", &array_Two[i]);
    }

    // Step 1: Sort array_One in Ascending order - O(N log N)
    qsort(array_One, n, sizeof(int), compare_asc);

    // Step 2: Sort array_Two in Descending order - O(N log N)
    qsort(array_Two, n, sizeof(int), compare_desc);
```

	Input	Expected	Got	
✔	3 1 2 3 4 5 6	28	28	✔
✔	4 7 5 1 2 1 3 4 1	22	22	✔
✔	5 20 10 30 10 40 8 9 4 3 10	590	590	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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